Numerical Verification Guide

Burman, Hansbo, Larson and Larsson

25 October 2026

This guide presents the detailed numerical results, accuracy checks and exact geometric certificates for *One-Sided p-Explicit Stability for High-Order Cut Finite Element Nitsche Methods*. The repository's `README.md` provides installation and quick-start instructions. Its linked `docs/reproduction.md` follows the study order below and gives commands, compact reference tables and the independent certificate verifier. The repository contains this PDF; manuscript sources are maintained separately.

The command `python report.py --published --output published` exports the named CSV reference tables to `published/published-tables/`. Detailed run records and exact certificates are regenerated by the experiment commands. Paths beginning with `output/` below are relative to the chosen experiment directory. The main paper's Appendix A summarizes the verification protocol and precision implementation. The 118-configuration campaign, retained accuracy flags, higher-degree comparisons, Schwarz results, flux spectra and geometric checks are included. The code and software documentation are licensed under MIT; numerical data and this guide are licensed under CC BY 4.0. Attribution: Erik Burman, Peter Hansbo, Mats G. Larson and Karl Larsson, *Numerical Companion to One-Sided p-Explicit Stability for High-Order Cut Finite Element Nitsche Methods*.

1 Numerical Results

1.1 Continuation Barrier

The main paper gives the compact continuation comparison in Figure 3 and the shared setup in Section 6. They are repeated here to keep the verification record self-contained. We first isolate the distinction between complete-element continuation and one-sided control. The numerically computed generalized-eigenvalue continuation constant C_p in (3.1) is computed in an orthonormal Legendre basis on I_-. Figure 1 compares it with the constructive Chebyshev lower bound from Proposition 3.1, the earlier monomial bound, and the endpoint constant $p+1$ from Lemma 4.1. The Chebyshev bound has the same sharp order as C_p, whereas the endpoint constant is linear in p. At $p = 14$, the recomputed values are $C_{14} = 9.57615 \times 10^9$ and $B_{14} = 8.14801 \times 10^9$, compared with the monomial bound 2.31705×10^4. Independent 60- and 80-digit evaluations agree to 10^{-50} relatively; direct integration also checks the closed formula for B_p.

1.2 Experimental Setting and Accuracy Criteria

The physical domain is the disk

$$\Omega_c = \{x : |x - c| < 0.62\} \subset (-1, 1)^2$$

The background square contains N^2 squares split into two affine triangles each, and $h = 2/N$. We use the conforming hierarchical $\mathbb{P}_p$ space with consistently oriented integrated-Legendre edge modes and factor-preserving interior bubbles. Appendix B defines this basis and explains why it spans $\mathbb{P}_p$, together with the conformity and evaluation checks. Boundary patches contain

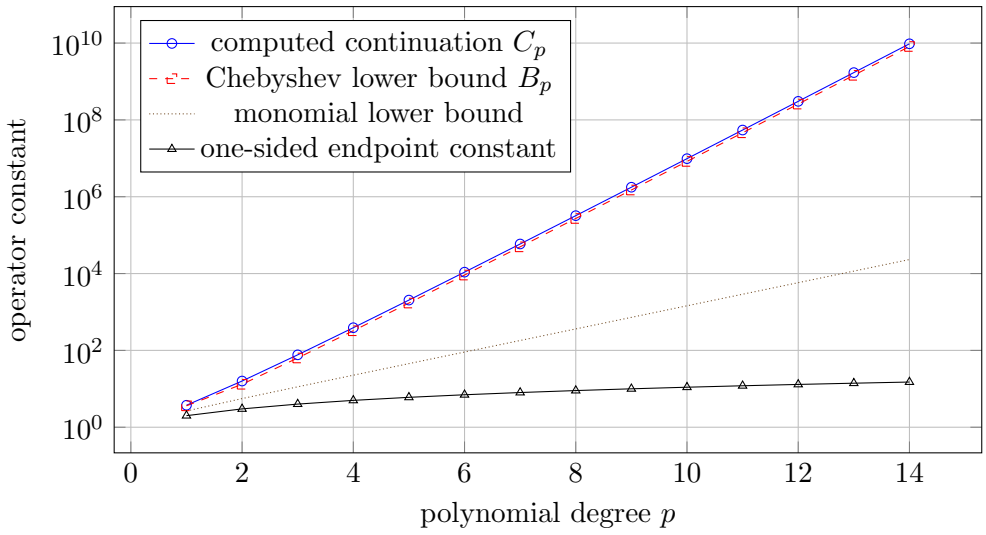


Figure 1: Exponential complete-element continuation versus polynomial one-sided endpoint control.

the complete active triangles within L face-adjacency steps of each boundary element. Unless stated otherwise, $L = 2$, $\gamma_1 = 0.1$, $\gamma_0 = 0$, and the Nitsche penalty is $16p^2/h$. Fixed centers are $c_A = (0.071, -0.043)$ and $c_B = (-0.113, 0.087)$.

The smooth manufactured solution is $u = \exp(x_1 + x_2)$, with $f = -2u$ and boundary data $g = u$. We also use $u = |x - c|^{5/2}$ with the corresponding Poisson load $f = -(25/4)|x - c|^{1/2}$ and boundary data $g = 0.62^{5/2}$. The ordinary manufactured loads are used throughout; no known-solution stabilization load is added. In each case, Eu is the same displayed formula evaluated on the whole active mesh, fixed independently of h and p. For this declared extension Eu and $e = Eu - u_h$, we measure the norm in (4.23) as

$$|||e|||_h^2 = \|\nabla e\|_\Omega^2 + \gamma_1 \sum_{\mathcal{P}} \|\nabla(e - E_{\mathcal{P}} e)\|_{\mathcal{P}}^2 + \frac{p^2}{h}\|e\|_{\partial\Omega}^2 \tag{1.1}$$

We save all three weighted squared contributions, their square roots and shares, and their element or patch contributions. The stabilization term acts on the error, including the extension on complete patches. As in the main paper, s_h includes the factor γ_1; the boundary weight in the norm excludes β_0. The L^2 error is reported separately; the gradient panels show the physical seminorm $\|\nabla e\|_\Omega$, rather than the full H^1 norm. For interpretation, define the physical part

$$E_{\mathrm{phys}}^2 = \|\nabla(u - u_h)\|_\Omega^2 + \frac{p^2}{h}\|u - u_h\|_{\partial\Omega}^2 \tag{1.2}$$

Changing Eu outside Ω can change the stabilization contribution to (1.1); it does not change u_h for the fixed method and physical load, E_{phys}, or the physical L^2 error. Stabilization shares refer to squared energy. A large share is therefore not a statement that the same fraction of physical error lies outside the domain. Physical errors and the full energy error are reported separately; selected comparisons appear in Table 18.

The practical baseline uses binary64 assembly and online field evaluation, compensated local accumulation, and QR projection residuals. The cached basis construction uses exact or extended arithmetic. Dense direct solves receive six binary64 correction solves with extended-precision residuals and working iterates; the returned coefficients are rounded to binary64. Independent error preparation uses Decimal arithmetic and extended-precision evaluation. These validation calculations are distinct from production assembly and evaluation. Neither the stability metrics nor the Schwarz local matrices receive diagonal shifts. The baseline local residual

Gram blocks are symmetrized, and eigenvalues below $2\times10^{-12}\max(1,\max_j|\lambda_j|)$ are set to zero. Diagnostics record the raw minimum eigenvalue, scale, clipping threshold, number of clipped modes and discarded absolute trace. This cleanup applies to the local stabilization blocks, not to the stability metrics or Schwarz local matrices.

The baseline campaign uses two assembly rules and two independently regenerated error rules for every PDE configuration. Smooth assembly boosts are 8/12 and error boosts 16/20; the radial counterparts are 12/16 and 20/24. A boost b increases the volume Gauss order to $\max(5,p+3+b)$ and the patch order to $\max(6,p+3+b)$, along with angular refinement; actual nodes and orders are retained. The circle is parametrized analytically, with quadrature split at mesh intersections. Strict active-cell classification is checked against higher-precision geometry. Basis conformity, area, QR, residual and integration checks accompany every run. A subsequent exact geometry audit checks all 30 distinct disk/patch configurations represented by the 118 standard PDE cases. The natural assignment to each boundary element's own seed patch fails the prescribed depth $h/16$ on 18 configurations. Reassignment among the existing overlapping patches certifies that depth for all 30 configurations: 97 of 5324 open angular pieces change their assigned patch. Every assigned patch contains the complete boundary element and all triangles traversed by its tube. The radial Jacobian is at least 1/2. Section A.5 describes the complete event partition and independent certificate verification. No patch enlargement, modified stabilization or new PDE solve is involved. This establishes the tube condition for the retained geometries; it does not prove a cut-uniform assignment algorithm on untested geometries or certify the sufficient analytic penalty threshold at the numerical choice $\beta_0=16$.

Paired assembly fields must differ by at most 1% of the smaller measured discretization error in each reported norm. Independent error rules must agree within 10^{-4} relatively; component differences are scaled by total squared energy. Selected smooth cases also have independent reference solutions, subject to the same 1% field-discrepancy criterion. Paired rules alone cannot exclude a shared bias. Unresolved points are retained as open markers and excluded from fits. The original campaign contains 102 distinct PDE configurations: 48 cut, 40 smooth and 14 radial cases. A subsequent mesh extension adds sixteen smooth configurations, giving 118 distinct standard configurations: 48 cut, 56 smooth (with eight overlaps between the mesh and degree grids), and 14 radial cases. The extension uses the unchanged assembly, solve, online field evaluation and accuracy rules. It computes no new coercivity spectra or penalty thresholds and has no independent reference solves for its new configurations. Its acceptance therefore denotes paired-rule agreement under the declared checks, rather than an independent accuracy certificate.

1.3 Cut Stability

At $N=6$ we take $p=2,4,6$ and three translation directions proportional to $(1,0)$, $(1,1)$ and $(1,2)$. A selected mesh vertex lies at distance $R-\epsilon h$ from the disk center, with $\epsilon\in\{3\times10^{-2},3\times10^{-3},3\times10^{-4},3\times10^{-5}\}$. The centers and mesh coordinates are frozen across degrees and layer counts. All 36 combinations use $L=2$; the twelve $p=4,6$ cases at the two endpoint offsets are also run with $L=4$. Cut severity is the measured minimum positive physical-cell fraction $\eta_{\min}=\min_{T\cap\Omega\neq\emptyset}|T\cap\Omega|/|T|$.

Writing K, S, M_∂ and C for the volume gradient, raw stabilization, boundary mass and consistency matrices, respectively, we use

$$E=K+\gamma_1 S+(p^2/h)M_\partial,\qquad A(\beta_0)=K+\gamma_1 S-C-C^T+\beta_0(p^2/h)M_\partial$$

Figure 2 shows cut sensitivity, and Table 1 summarizes the group extrema. The reported threshold brackets enclose the crossing of $\lambda_{\min}(A(\beta_0),E)=10^{-8}$. Their widths are at most $10^{-4}\max(1,\beta_0)$, with checked endpoint signs, eigenpair residuals, unshifted metric transforms, and 10^{-3} paired-rule threshold agreement. All 48 cases pass the declared checks. The smallest

measured fraction is 7.50×10^{-10}; threshold brackets lie within $[0.4839, 1.9132]$. At $\beta_0 = 16$, the minimum generalized eigenvalues range from 0.8672 to 0.9683 over all 48 cases. These measurements support bounded thresholds on the sampled paths and offsets; they do not establish a numerical bound for arbitrary tangencies.

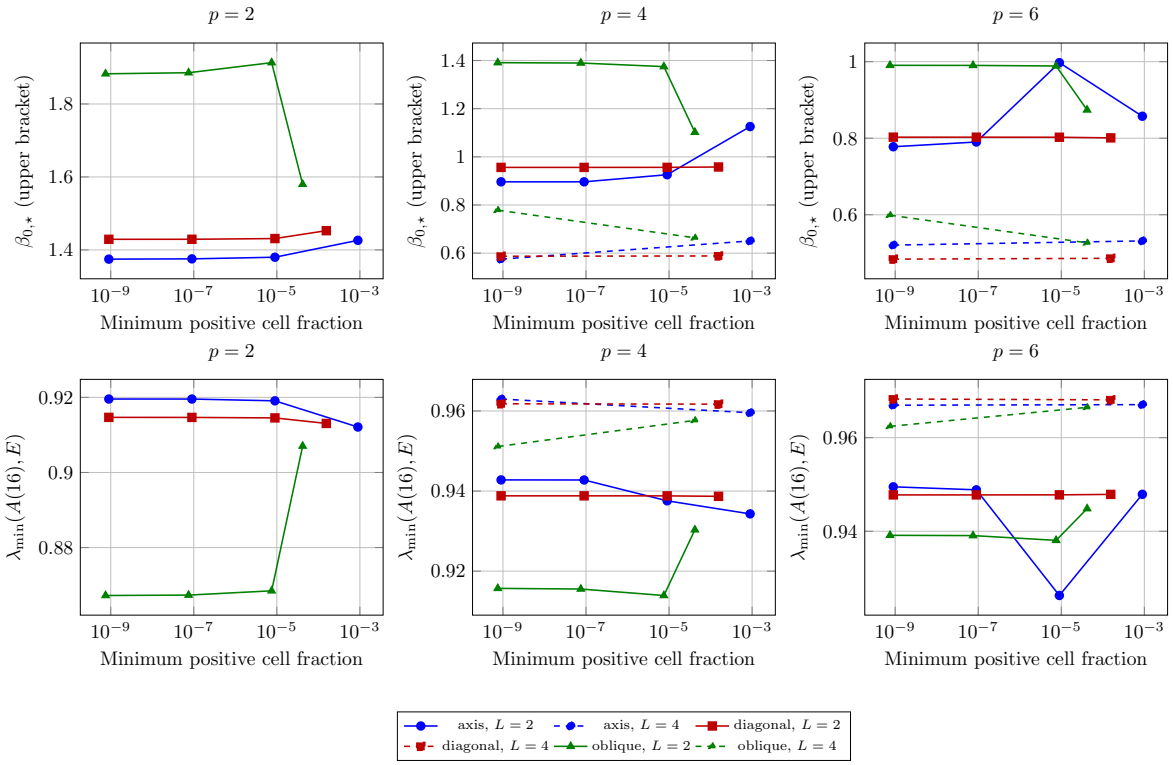


Figure 2: Checked threshold upper endpoints and nominal coercivity versus actual cut fraction. Colors identify translation directions; solid and dashed curves use stabilization depths $L = 2$ and $L = 4$.

1.4 Smooth Mesh and Degree Convergence

For both fixed disks, the extended mesh study uses $N = 4, 6, 8, 12, 16, 24$ and $p = 2, 4, 6, 8$; the degree study remains $N = 6$, $p = 1, \ldots, 8$. The 48 mesh configurations include 32 saved configurations and sixteen new ones. All sixteen new calculations complete, but only six pass every accuracy criterion: $N = 24$ at $p = 2, 4$, and $N = 4$ at $p = 8$, on both disks. Degree six remains resolved through $N = 16$ and fails all three 1% paired-field criteria at $N = 24$. Degree eight is resolved at $N = 4$ in every norm and at $N = 6$ in energy only; its $N = 8, 12, 16, 24$ values fail all three norm criteria. The mesh study thus contains 36 resolved L^2 and gradient values and 38 resolved energy values. Every unresolved measurement remains visible.

Table 2 fits the finest three consecutive resolved levels for each family. The degree-two and degree-four windows now reach $N = 24$; degree six retains $N = 8, 12, 16$. No degree-eight family has three consecutive resolved levels, so no validated mesh fit is reported at that degree. Full eligible segments and local slopes, including raw slopes between flagged measurements, are retained in the data. For degree six, the L^2 slope remains 6.50–6.52 over the five resolved meshes and 7.19–7.55 over the finest three. These are observed finite-grid rates, not stronger asymptotic orders.

At the fixed mesh $N = 6$, the degree curves are resolved through $p = 7$ in all three norms and through $p = 8$ in energy. At $p = 8$, the independent-reference field discrepancies are 1.33%/1.58% in L^2, 1.79%/2.10% in the gradient seminorm, and 0.358%/0.423% in energy for

Table 1: Cut-study extrema within each sampled group. Patch diameters, overlap counts and individual brackets are retained in the accompanying data.

Path	p	L	$\eta_{\min}$	Max. threshold upper	Min. coercivity
axis	2	2	9.00×10^{-10}	1.4263	0.9121
axis	4	2	9.00×10^{-10}	1.1257	0.9343
axis	6	2	9.00×10^{-10}	0.9974	0.9263
axis	4	4	9.00×10^{-10}	0.6508	0.9596
axis	6	4	9.00×10^{-10}	0.5316	0.9669
diagonal	2	2	9.00×10^{-10}	1.4528	0.9131
diagonal	4	2	9.00×10^{-10}	0.9578	0.9387
diagonal	6	2	9.00×10^{-10}	0.8027	0.9478
diagonal	4	4	9.00×10^{-10}	0.5883	0.9617
diagonal	6	4	9.00×10^{-10}	0.4861	0.9681
oblique	2	2	7.50×10^{-10}	1.9132	0.8672
oblique	4	2	7.50×10^{-10}	1.3911	0.9139
oblique	6	2	7.50×10^{-10}	0.9905	0.9381
oblique	4	4	7.50×10^{-10}	0.7786	0.9511
oblique	6	4	7.50×10^{-10}	0.5989	0.9624

Table 2: Measured mesh slopes fitted independently in each norm over the indicated finest three consecutive resolved levels. Flagged points do not enter any fit.

Disk	p	Resolved N window	L^2	Gradient	Energy
A	2	12,16,24	3.332	2.079	2.542
A	4	12,16,24	5.455	4.510	4.684
A	6	8,12,16	7.546	6.709	6.708
B	2	12,16,24	3.435	2.124	2.512
B	4	12,16,24	5.695	4.704	4.686
B	6	8,12,16	7.192	6.463	6.632

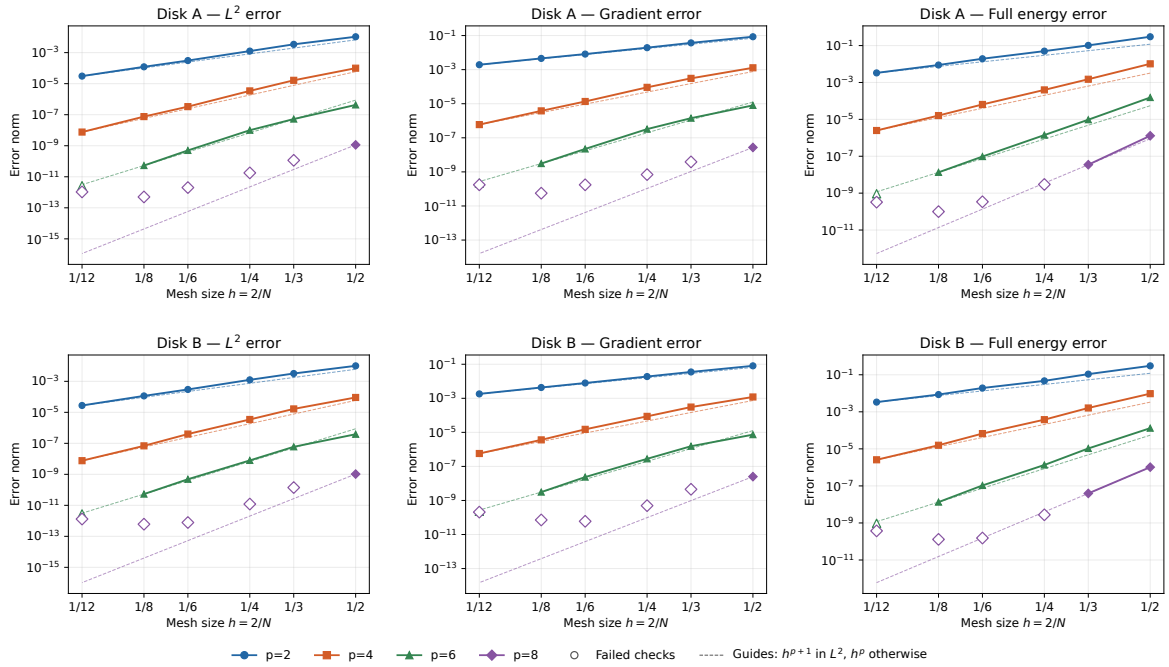


Figure 3: Extended smooth mesh convergence on disks A and B through $N = 24$ and $p = 8$. Filled markers pass the norm-specific checks; open markers retain flagged values, and connected curves do not bridge them. Dashed guides have powers $p+1$ in L^2 and p in gradient and energy, normalized at the finest resolved point. They are reference powers, not error bounds.

disks A/B. Thus the separate L^2 and gradient criteria fail at $p = 8$, even though the full-energy criterion passes. This energy acceptance at $N = 6$ does not extend to finer degree-eight meshes. At $N = 16, p = 6$, the L^2 errors are about 5.3×10^{-11} and energy errors about 1.34×10^{-8}; stabilization contributes 94.6–94.7% of squared energy. The component data explain why a small full-energy discrepancy can coexist with a larger relative discrepancy in a physical norm.

At $N = 24, p = 6$, the assembly-field differences are 3.12–3.41% in L^2, 4.38–5.48% in gradient and 1.85–2.25% in energy. At degree eight, refining $N = 16$ to $N = 24$ increases the measured L^2, gradient and energy errors by factors 2.11, 3.11 and 3.26 on disk A, and 2.15, 2.89 and 2.87 on disk B. Both endpoints are flagged, so this reversal describes computed fields rather than a validated convergence rate. Norm integration repeats remain within 10^{-4}; some fine degree-eight solves also fail the retained last-correction field checks. Appendix A.4 gives the finest-mesh measurements, all ten new paired-field failures and the extended energy components. These observations delimit the standard implementation's accuracy range. Without new independent references or spectra, they neither isolate every source of arithmetic sensitivity nor provide a counterexample to the stability theorem.

Figure 5 separates the weighted energy contributions for mesh and degree refinement.

1.5 Higher-Degree Accuracy and Precision Limits

We supplement the standard campaign with two degree-eight accuracy anchors on disks A and B and degree-ten and degree-twelve comparisons on disk A. These use $N = 6$, $L = 2$ and the same space, equation, ordinary exponential data and stabilization parameters. The original degree-eight flags in Figure 4 remain unchanged. These fixed-$N = 6$ precision comparisons do not validate the finer-mesh points in Figure 3.

The accurate path uses analytic circle moments and patch projection complements at 80 decimal digits, local operator contractions at 100 digits, and 24 corrections with 100-digit residuals

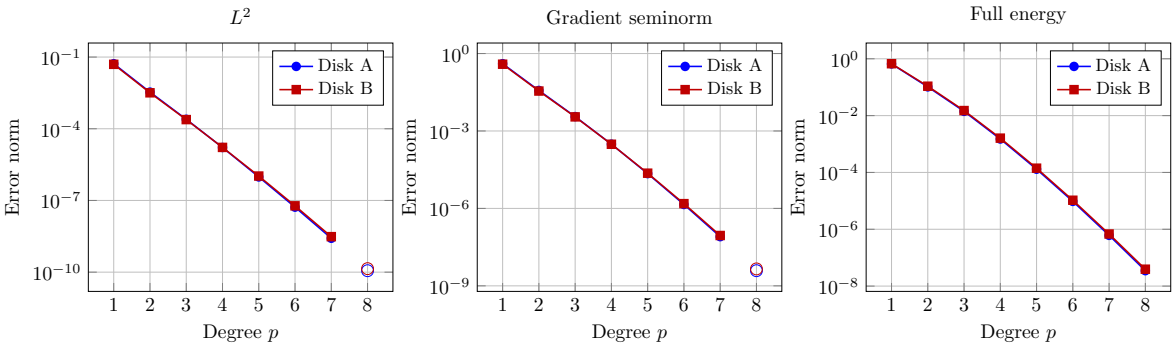


Figure 4: Smooth degree convergence at $N = 6$. Open degree-eight L^2 and gradient markers retain the measured values that fail the 1% independent-reference criterion.

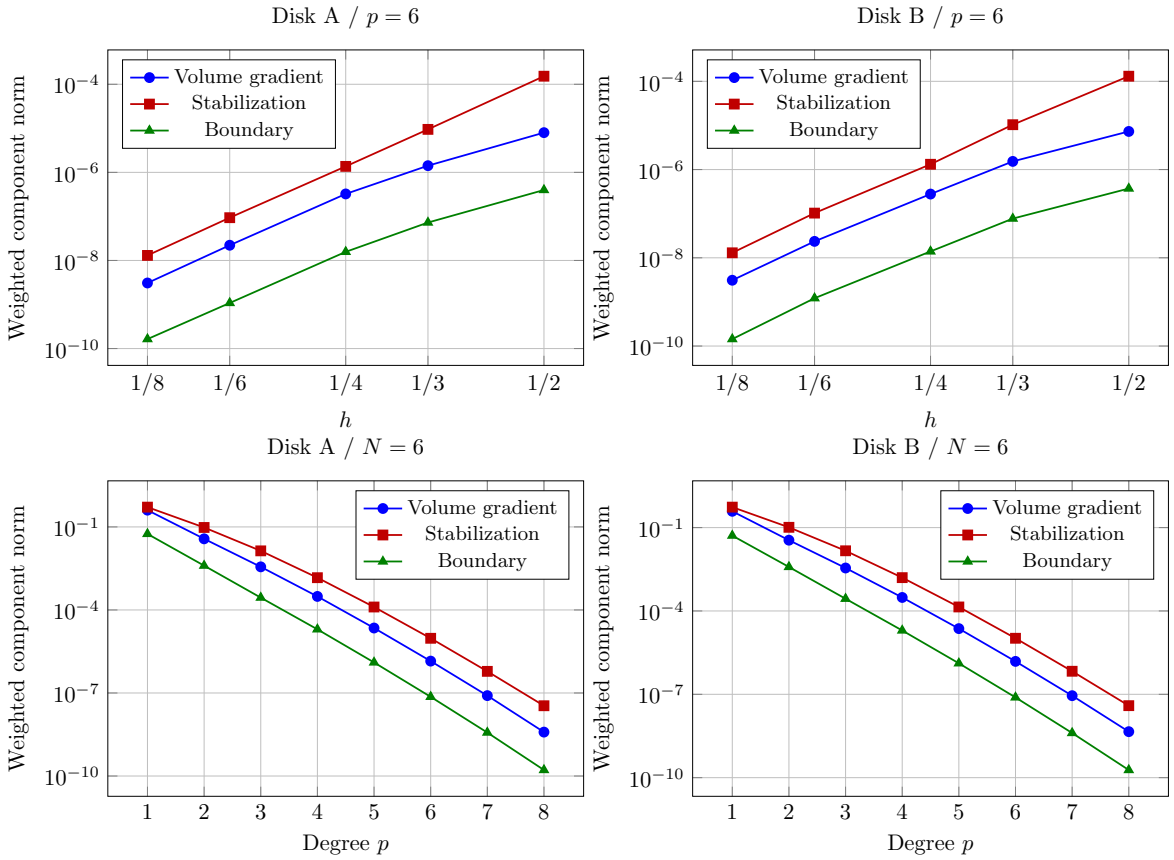


Figure 5: Square roots of the weighted energy contributions: the original degree-six mesh study through $N = 16$ (top) and degree refinement at $N = 6$ (bottom). The extended degree-six/eight mesh components appear in Appendix A.4. Their squares, rather than the displayed square roots, add to squared energy.

and coefficient updates. One diagonally scaled binary64 LU factorization supplies the correction solves. Global stabilization is accumulated from the same local projection complements without dropping small entries. Exported coefficients and Bernstein controls retain separate binary64 high and low components; online evaluation uses paired binary64 operations. No known-solution stabilization load or diagonal shift is added. The ordinary comparison uses the saved degree-eight binary64 coefficients and fresh boost-8/12 assemblies with six corrections at degrees ten and twelve.

Independent coefficient-field error identities give the norms in Table 3 and Figure 6. The accurate fields meet the unchanged 1% reference-field criterion in all three norms, and error boosts 16/20 pass their 10^{-4} repeat checks. Reference coefficients enter validation only; fresh assembly reuses the established analytic implementation underlying those references. Paired online arithmetic is checked separately at six physical points on each active triangle, against a 100-digit power-basis oracle, with maximum discrepancy below 2×10^{-31}. Full-triangle coefficient-storage and control-export bounds are retained. These norm measurements are not integrals of the online paired arithmetic at every quadrature point.

Table 3: Accurate coefficient-field errors at $N = 6$, $L = 2$. The degree-ten and degree-twelve results cover disk A only.

Disk	p	L^2	Gradient	Full energy
A	8	1.15×10^{-10}	3.87×10^{-9}	3.48×10^{-8}
B	8	1.40×10^{-10}	4.53×10^{-9}	3.93×10^{-8}
A	10	1.55×10^{-13}	6.05×10^{-12}	8.21×10^{-11}
A	12	1.44×10^{-16}	6.42×10^{-15}	1.36×10^{-13}

The ordinary path loses physical accuracy at higher degree, including in full energy. At $p = 12$, its gradient and energy errors are 4.80×10^{-9} and 8.72×10^{-9}, versus 6.42×10^{-15} and 1.36×10^{-13} for the accurate path. At $p = 10$, ordinary assembly-rule gradient-field differences are about 255 times the reference discretization error despite corrected stored-system backward errors around 6×10^{-20}. Accurate assembly recovers the degree decay, indicating a substantial accuracy limitation before the linear solve. Collapsing the accurate degree-twelve coefficients to one binary64 array produces reference-field discrepancies of 25.9%, 6.13% and 1.45% in L^2, gradient and energy, respectively. All exceed the 1% target. Three additional ordinary stabilization-component repeat flags remain recorded in Appendix A.3; the complete degree-twelve comparison is not marked wholly accepted.

The accurate path is a dense research implementation with additional precision costs. For the 2467-DOF degree-twelve case, the recorded assembly phase takes 207 seconds on a four-CPU allocation, including saved assembly output. The two load-series correction phases total 31.7 seconds, excluding the reused LU factorization, setup, export and independent validation. These timings are descriptive and separate from Table 5. No new coercivity spectra or penalty thresholds are computed. This finite comparison supports continued degree decay of the accurately computed discrete method, without establishing uniform geometric robustness or a scalable solver.

1.6 Finite Regularity in the Full Energy Norm

At $N = 4$, both fixed disks use $u = |x-c|^{5/2}$ and $p = 2,\dots,8$. This extension has H^s regularity for $s < 7/2$. For each fixed $0 < \varepsilon < 1/2$, the real-index versions of Theorems 5.4 and 5.6 give fixed-mesh bounds $O(p^{-2+\varepsilon})$ in energy and $O(p^{-5/2+\varepsilon})$ in L^2. The plotted powers p^{-2} and $p^{-5/2}$ are their limiting guides; neither endpoint regularity nor uniformity as $\varepsilon \downarrow 0$ is asserted. All fourteen cases pass paired assembly, independent norm integration, component, geometry and solver checks. The declared primary fits use $p = 4,\dots,8$ and sensitivity fits use $p = 3,\dots,7$.

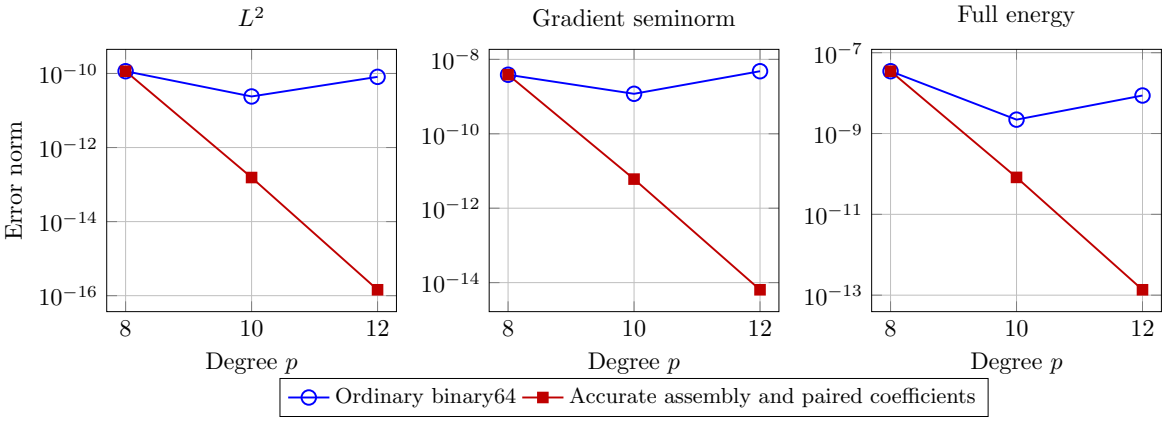


Figure 6: Disk A, $N = 6$, $L = 2$: coefficient-field error decay with accurate assembly and paired coefficients, compared with the ordinary binary64 implementation. Circles denote ordinary coefficients and squares the accurate path. The ordinary degree-eight energy criterion passes, while the degree-ten and degree-twelve ordinary comparisons fail all three norm criteria. Online paired evaluation is probed separately.

Table 4: Finite-regularity degree slopes. Differences between windows quantify the sensitivity of the fitted exponent. Rates are observations, not acceptance tests.

Disk	Norm	$p = 4, \ldots, 8$	$p = 3, \ldots, 7$
A	L^2	-3.189	-3.668
A	Gradient	-2.501	-2.911
A	Energy	-2.599	-2.765
B	L^2	-3.210	-3.686
B	Gradient	-2.518	-2.921
B	Energy	-2.646	-2.859

Table 4 reports the fitted slopes, and Figure 7 displays the error curves. The primary energy slopes are -2.646 to -2.599. At $p = 8$, stabilization contributes 89.3–90.3% of squared energy. The observed finite-grid decay is faster than the limiting guaranteed guides, but the sensitivity window does not support identifying a unique asymptotic exponent from these seven degrees alone.

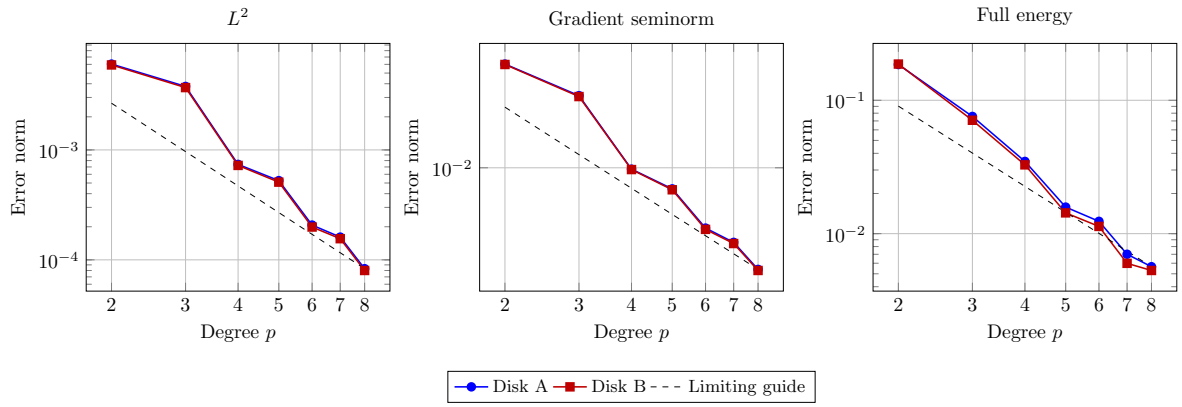


Figure 7: Finite-regularity degree convergence in the physical and full energy norms. Dashed lines show limiting comparison powers, normalized at disk A, $p = 8$. The estimates retain an arbitrarily small exponent loss because $s < 7/2$; the lines are not error bounds.

The separate finite-regularity energy components appear in Appendix A.

1.7 A Compact Schwarz Comparison

The algebraic comparison reuses the saved higher-order smooth operators, with stabilization depth fixed at $L = 2$. It comprises six smallest-offset cut operators ($p = 2, 4, 6$, axis and diagonal paths), with preconditioner depths $L_{\mathrm{pc}} = 1, 2, 4$, and the ten degree-four mesh-study operators with $L_{\mathrm{pc}} = 4$. Each of these sixteen operators also has a Jacobi comparison: 28 Schwarz and 16 Jacobi applications in total.

We work in diagonally scaled coordinates. Duplicate patch subdomains are removed, local matrices use unshifted scaled Cholesky factors, and symmetric inverse-square-root overlap weights form the additive preconditioner. Uncovered interior degrees of freedom receive an exact global block solve. We check the endpoint eigenpairs of the symmetric preconditioned operator and retain setup, application and iteration times. This dense construction and the global interior block preclude a scalable-solver claim.

PCG starts at zero. We retain the first checked true-residual checkpoint below 10^{-9} and continue until the saved-metric field discrepancy is at most 10^{-3} of the direct discretization energy error. This second stopping test uses the direct solution for validation; it is not a reference-free stopping rule. Independent integration then requires at most 1% discrepancy in L^2 and gradient, and 0.1% in energy. One $N = 16, p = 4$, disk-A Schwarz result initially failed the L^2 test at 1.54%. Its retained follow-up tightens only the energy-relative stopping target to 10^{-5} for both preconditioners on that operator, retaining the original acceptance criteria and all first-attempt data. The Schwarz follow-up passes with 69 iterations and a 0.0088% L^2 discrepancy; all 44 selected comparisons pass their declared field and spectral checks. Individual cut-operator comparisons are retained in Table 14.

For the ten fixed-disk degree-four operators, four-layer condition estimates range from 7.19 to 54.30, with 20–69 validated PCG iterations. The cut-operator degree-six condition estimates remain about 3.6×10^3 and 4.0×10^4 even with four layers. The observed improvement over

Jacobi therefore does not establish p-robustness. Since these comparisons sample only the smallest offset on two paths, the new solver study alone does not establish uniform cut robustness. Figure 9 in Appendix A shows the condition estimates.

1.8 Practical Accuracy, Cost and Reproducibility

In the original 102-case campaign, the practical baseline resolves all 48 cut and 14 radial configurations and 38 of the 40 smooth configurations under every declared criterion. The two smooth degree-eight failures remain visible; all 102 full-energy results pass their declared checks. Extended references validate selected accuracy anchors in that campaign. The sixteen subsequent mesh configurations add six wholly accepted results and ten results flagged in every norm. Across the 118 standard configurations, 106 pass every criterion and 108 pass the energy criteria. The four separate higher-precision comparisons are not included in these standard-case counts. The additional accurate assembly path and its precision costs are reported separately in Section 1.5. The data retain separate accuracy and stability status, source hashes, parameters, package versions, quadrature identities, scheduler completion records, global and local energy components, and unsuccessful checks.

Table 5: Representative disk-A operation times. Setup includes mesh, space and cached basis preparation; solve includes the six corrections. Independent integration, basis validation and spectral checks are excluded. The dagger marks the degree-eight row whose separate L^2/gradient accuracy criteria fail; its energy criterion passes.

N	p	DOFs	CPUs	Setup (s)	Assembly (s)	Solve (s)
6	4	295	2	0.01	6.87	0.0049
6	6	640	2	0.02	11.84	0.0411
6	$8^{\dagger}$	1117	2	0.07	14.87	0.1939
16	6	3550	6	0.04	59.02	3.6842

The workstation has heterogeneous CPU cores. Timings with different allocations are descriptive and do not establish parallel speedup. Whole-job time includes tests, source checks, independent error integration and expensive spectral validation in the original campaign; it is not production solve time. The new convergence jobs omit spectra and threshold searches, so their whole-job times are not comparable with the original validation jobs. Their largest degree-eight systems at $N = 24$ have 12853 and 13017 degrees of freedom on disks A and B. Separate assembly, initial solve, correction and norm-integration timers, CPU allocations and measured peak memory remain in the saved data. The inherited spectral timer ends before its full extended-precision checks and is labelled as partial in the data. GPU acceleration and a scalable preconditioner were not tested.

2 Additional Experiments Used in the Main Paper

The records below supplement the original campaign; all use the same finite element formulation unless explicitly marked as a fitted or rescaled control. The reproduction repository includes the exact machine-readable inputs in `data/cases.json` and reference outcomes in `data/published.json.gz`, including unresolved values and retained accuracy flags.

2.1 Direct Flux Spectra and Matched Thresholds

Tables 6, 7 and 8 give all 96 degree/placement combinations in main-paper Figure 4. The columns contain C_1^{num}, $C_{0.1}^{\mathrm{num}}$ and the unstabilized individual-basis witness $b_0 \le C_0^{\mathrm{num}}$. The raw

records also retain all attempted unstabilized eigenvalues and flags: 26 pass and 70 remain unresolved. All 192 stabilized spectra pass. Pinning a vertex removes only the constant nullspace; no shift or eigenvalue clipping is applied. Paired rules must agree within 10^{-4}, and whitened-metric and eigenpair checks must be below 10^{-5}. The data file `direct-flux.csv` retains the values, flags and repeat diagnostics. Each `flux` run writes both rules and per-eigenpair residuals to `output/cut-results.json`. Table entries are rounded for display; computations and comparisons use the saved values. The threshold columns reproduce the original $p = 2, 4, 6$, $L = 2$ brackets; a dash marks a degree with no threshold measurement. All 36 upper endpoints are below their matched computational flux constants, with smallest observed margin 7.79×10^{-4}.

Table 6: All axis-path direct flux measurements at $N = 6$, $L = 2$. Witnesses are lower bounds, not resolved unstabilized eigenvalues.

ϵ	p	$\eta_{\min}$	C_1^{num}	$C_{0.1}^{\mathrm{num}}$	b_0	Threshold upper
3×10^{-2}	1	8.95×10^{-4}	0.861622	2.410962	4.765×10^{1}	–
3×10^{-2}	2	8.95×10^{-4}	0.713884	1.537405	2.064×10^{1}	1.4263
3×10^{-2}	3	8.95×10^{-4}	0.625308	1.376206	1.262×10^{1}	–
3×10^{-2}	4	8.95×10^{-4}	0.592750	1.195482	9.023×10^{0}	1.1257
3×10^{-2}	5	8.95×10^{-4}	0.589720	1.014118	7.014×10^{0}	–
3×10^{-2}	6	8.95×10^{-4}	0.578007	0.884621	5.742×10^{0}	0.8574
3×10^{-2}	7	8.95×10^{-4}	0.573186	0.811819	4.866×10^{0}	–
3×10^{-2}	8	8.95×10^{-4}	0.563399	0.775073	4.226×10^{0}	–
3×10^{-3}	1	9.00×10^{-6}	0.864719	2.357317	4.928×10^{2}	–
3×10^{-3}	2	9.00×10^{-6}	0.707065	1.387015	2.081×10^{2}	1.3800
3×10^{-3}	3	9.00×10^{-6}	0.623689	1.066402	1.288×10^{2}	–
3×10^{-3}	4	9.00×10^{-6}	0.590724	1.070602	9.277×10^{1}	0.9257
3×10^{-3}	5	9.00×10^{-6}	0.586384	1.165814	7.238×10^{1}	–
3×10^{-3}	6	9.00×10^{-6}	0.575194	1.242001	5.930×10^{1}	0.9974
3×10^{-3}	7	9.00×10^{-6}	0.569134	1.287842	5.022×10^{1}	–
3×10^{-3}	8	9.00×10^{-6}	0.560220	1.304237	4.354×10^{1}	–
3×10^{-4}	1	9.00×10^{-8}	0.866491	2.351976	4.977×10^{3}	–
3×10^{-4}	2	9.00×10^{-8}	0.706557	1.380363	2.083×10^{3}	1.3755
3×10^{-4}	3	9.00×10^{-8}	0.623495	1.039349	1.294×10^{3}	–
3×10^{-4}	4	9.00×10^{-8}	0.590494	0.897019	9.345×10^{2}	0.8962
3×10^{-4}	5	9.00×10^{-8}	0.586011	0.832656	7.304×10^{2}	–
3×10^{-4}	6	9.00×10^{-8}	0.574939	0.821976	5.990×10^{2}	0.7901
3×10^{-4}	7	9.00×10^{-8}	0.568707	0.850233	5.076×10^{2}	–
3×10^{-4}	8	9.00×10^{-8}	0.559931	0.899787	4.403×10^{2}	–
3×10^{-5}	1	9.00×10^{-10}	0.867119	2.351442	4.993×10^{4}	–
3×10^{-5}	2	9.00×10^{-10}	0.706623	1.379594	2.083×10^{4}	1.3748
3×10^{-5}	3	9.00×10^{-10}	0.623475	1.039158	1.296×10^{4}	–
3×10^{-5}	4	9.00×10^{-10}	0.590477	0.897054	9.366×10^{3}	0.8962
3×10^{-5}	5	9.00×10^{-10}	0.585974	0.824982	7.324×10^{3}	–
3×10^{-5}	6	9.00×10^{-10}	0.574913	0.779595	6.010×10^{3}	0.7777
3×10^{-5}	7	9.00×10^{-10}	0.568665	0.755843	5.094×10^{3}	–
3×10^{-5}	8	9.00×10^{-10}	0.559902	0.748167	4.419×10^{3}	–

Table 7: All diagonal-path direct flux measurements at $N = 6$, $L = 2$. Witnesses are lower bounds, not resolved unstabilized eigenvalues.

ϵ	p	$\eta_{\min}$	C_1^{num}	$C_{0.1}^{\mathrm{num}}$	b_0	Threshold upper
3×10^{-2}	1	1.55×10^{-4}	0.881530	2.450601	9.263×10^{1}	–
3×10^{-2}	2	1.55×10^{-4}	0.749332	1.482262	5.928×10^{1}	1.4528
3×10^{-2}	3	1.55×10^{-4}	0.640335	1.099962	4.080×10^{1}	–
3×10^{-2}	4	1.55×10^{-4}	0.614414	0.962320	3.067×10^{1}	0.9578
3×10^{-2}	5	1.55×10^{-4}	0.594772	0.875834	2.448×10^{1}	–
3×10^{-2}	6	1.55×10^{-4}	0.582612	0.802830	2.034×10^{1}	0.8009
3×10^{-2}	7	1.55×10^{-4}	0.577729	0.761745	1.739×10^{1}	–
3×10^{-2}	8	1.55×10^{-4}	0.567415	0.734597	1.519×10^{1}	–
3×10^{-3}	1	9.00×10^{-6}	0.882287	2.441179	3.327×10^{2}	–
3×10^{-3}	2	9.00×10^{-6}	0.747305	1.454255	1.663×10^{2}	1.4312
3×10^{-3}	3	9.00×10^{-6}	0.639888	1.098143	1.108×10^{2}	–
3×10^{-3}	4	9.00×10^{-6}	0.614484	0.960174	8.304×10^{1}	0.9563
3×10^{-3}	5	9.00×10^{-6}	0.595181	0.875654	6.643×10^{1}	–
3×10^{-3}	6	9.00×10^{-6}	0.582386	0.804353	5.535×10^{1}	0.8026
3×10^{-3}	7	9.00×10^{-6}	0.577387	0.763400	4.744×10^{1}	–
3×10^{-3}	8	9.00×10^{-6}	0.566768	0.735849	4.151×10^{1}	–
3×10^{-4}	1	9.00×10^{-8}	0.882367	2.440197	3.333×10^{3}	–
3×10^{-4}	2	9.00×10^{-8}	0.747122	1.451824	1.666×10^{3}	1.4292
3×10^{-4}	3	9.00×10^{-8}	0.639851	1.098076	1.111×10^{3}	–
3×10^{-4}	4	9.00×10^{-8}	0.614495	0.959937	8.330×10^{2}	0.9561
3×10^{-4}	5	9.00×10^{-8}	0.595218	0.875653	6.664×10^{2}	–
3×10^{-4}	6	9.00×10^{-8}	0.582364	0.804527	5.554×10^{2}	0.8027
3×10^{-4}	7	9.00×10^{-8}	0.577351	0.763574	4.760×10^{2}	–
3×10^{-4}	8	9.00×10^{-8}	0.566705	0.735971	4.165×10^{2}	–
3×10^{-5}	1	9.00×10^{-10}	0.882375	2.440099	3.333×10^{4}	–
3×10^{-5}	2	9.00×10^{-10}	0.747104	1.451584	1.667×10^{4}	1.4291
3×10^{-5}	3	9.00×10^{-10}	0.639848	1.098070	1.111×10^{4}	–
3×10^{-5}	4	9.00×10^{-10}	0.614497	0.959913	8.333×10^{3}	0.9561
3×10^{-5}	5	9.00×10^{-10}	0.595221	0.875653	6.666×10^{3}	–
3×10^{-5}	6	9.00×10^{-10}	0.582362	0.804545	5.555×10^{3}	0.8027
3×10^{-5}	7	9.00×10^{-10}	0.577347	0.763591	4.762×10^{3}	–
3×10^{-5}	8	9.00×10^{-10}	0.566699	0.735983	4.167×10^{3}	–

Table 8: All oblique-path direct flux measurements at $N = 6$, $L = 2$. Witnesses are lower bounds, not resolved unstabilized eigenvalues.

ϵ	p	$\eta_{\min}$	C_1^{num}	$C_{0.1}^{\mathrm{num}}$	b_0	Threshold upper
3×10^{-2}	1	4.18×10^{-5}	0.884347	2.788739	3.029×10^{2}	–
3×10^{-2}	2	4.18×10^{-5}	0.737708	1.899168	1.655×10^{2}	1.5800
3×10^{-2}	3	4.18×10^{-5}	0.645107	1.465721	1.106×10^{2}	–
3×10^{-2}	4	4.18×10^{-5}	0.609702	1.147950	8.274×10^{1}	1.1020
3×10^{-2}	5	4.18×10^{-5}	0.596974	0.976374	6.600×10^{1}	–
3×10^{-2}	6	4.18×10^{-5}	0.585367	0.904170	5.487×10^{1}	0.8739
3×10^{-2}	7	4.18×10^{-5}	0.576151	0.867664	4.695×10^{1}	–
3×10^{-2}	8	4.18×10^{-5}	0.567800	0.841144	4.102×10^{1}	–
3×10^{-3}	1	7.50×10^{-6}	0.891617	3.741383	3.986×10^{2}	–
3×10^{-3}	2	7.50×10^{-6}	0.742324	2.569445	2.571×10^{2}	1.9132
3×10^{-3}	3	7.50×10^{-6}	0.653096	1.995324	1.785×10^{2}	–
3×10^{-3}	4	7.50×10^{-6}	0.613002	1.545049	1.346×10^{2}	1.3748
3×10^{-3}	5	7.50×10^{-6}	0.593606	1.271927	1.074×10^{2}	–
3×10^{-3}	6	7.50×10^{-6}	0.580470	1.138180	8.928×10^{1}	0.9888
3×10^{-3}	7	7.50×10^{-6}	0.571775	1.060555	7.635×10^{1}	–
3×10^{-3}	8	7.50×10^{-6}	0.564353	0.999928	6.670×10^{1}	–
3×10^{-4}	1	7.50×10^{-8}	0.890645	3.683429	3.999×10^{3}	–
3×10^{-4}	2	7.50×10^{-8}	0.741587	2.556607	2.582×10^{3}	1.8856
3×10^{-4}	3	7.50×10^{-8}	0.652011	2.007199	1.794×10^{3}	–
3×10^{-4}	4	7.50×10^{-8}	0.612607	1.561180	1.352×10^{3}	1.3896
3×10^{-4}	5	7.50×10^{-8}	0.593435	1.273871	1.080×10^{3}	–
3×10^{-4}	6	7.50×10^{-8}	0.580305	1.126851	8.976×10^{2}	0.9902
3×10^{-4}	7	7.50×10^{-8}	0.571579	1.043236	7.677×10^{2}	–
3×10^{-4}	8	7.50×10^{-8}	0.564139	0.980069	6.707×10^{2}	–
3×10^{-5}	1	7.50×10^{-10}	0.890549	3.677498	4.000×10^{4}	–
3×10^{-5}	2	7.50×10^{-10}	0.741514	2.555201	2.583×10^{4}	1.8827
3×10^{-5}	3	7.50×10^{-10}	0.651900	2.008333	1.794×10^{4}	–
3×10^{-5}	4	7.50×10^{-10}	0.612566	1.562869	1.353×10^{4}	1.3911
3×10^{-5}	5	7.50×10^{-10}	0.593419	1.274248	1.081×10^{4}	–
3×10^{-5}	6	7.50×10^{-10}	0.580290	1.125895	8.981×10^{3}	0.9905
3×10^{-5}	7	7.50×10^{-10}	0.571560	1.041712	7.681×10^{3}	–
3×10^{-5}	8	7.50×10^{-10}	0.564118	0.978388	6.710×10^{3}	–

2.2 Fitted Polygon and Extension Seminorms

The fitted control has 25 vertices and 28 affine triangles. Its 20 boundary vertices are

$$b_j = 0.62(\cos(2\pi j/20), \sin(2\pi j/20)) \qquad j = 0, \ldots, 19$$

The five interior vertices are $(0, 0)$ and $(\pm 1/3, 0), (0, \pm 1/3)$, up to the stored binary64 rounding. The exact stored coordinates and triangle connectivity are in `data/fitted-geometry.json`. The nominal scale in the penalty is $h = 1/3$; actual element diameters range from 0.320012 to 0.502553, with maximum diameter divided by h equal to 1.507660. There is no stabilization and the boundary is the polygon, not the exact circle. Table 9 gives all fitted values used in main-paper Figure 4 and Tables 1 and 4. Paired quadrature orders are $p + 3$ and $p + 7$; flux values agree within 10^{-5}, condition/coercivity values within 10^{-4}, and threshold endpoints within 2×10^{-4}.

Table 10 gives all extension seminorms supporting main-paper Figure 8. These are $\|Eu\|_{s_h}$, not the error seminorm; the full original error components remain in `smooth-case-summary.csv`. Paired patch rules agree within 10^{-5}; each `controls` run writes the paired records to `output/extension-results.json`.

Table 9: Unstabilized fitted 20-gon control. Brackets locate the 10^{-8} energy-normalized crossing; coercivity and Jacobi conditioning use $\beta_0 = 16$.

p	DoFs	C_0^{num}	Lower	Upper	Coercivity	κ_J
1	25	1.769585	1.746765	1.746826	0.895434	3.6521
2	77	1.529075	1.523132	1.523193	0.903685	16.8742
3	157	1.437153	1.435364	1.435425	0.907516	33.6669
4	265	1.384621	1.383911	1.383972	0.910281	60.8296
5	401	1.350836	1.350525	1.350586	0.912097	76.9563
6	565	1.327166	1.326965	1.327026	0.913375	132.6169
7	757	1.309534	1.309387	1.309448	0.914321	158.5320
8	977	1.295807	1.295715	1.295776	0.915057	253.3966

Table 10: Extension seminorms at $N = 6$, $L = 2$, $\gamma_1 = 0.1$. Repeat columns are relative differences between patch rules.

p	A: $\|Eu\|_{s_h}$	B: $\|Eu\|_{s_h}$	A repeat	B repeat
1	6.48071×10^{-1}	6.60623×10^{-1}	2.2×10^{-16}	0
2	1.18780×10^{-1}	1.21920×10^{-1}	2.2×10^{-16}	2.2×10^{-16}
3	1.60686×10^{-2}	1.67951×10^{-2}	2.2×10^{-15}	1.8×10^{-15}
4	1.65953×10^{-3}	1.75996×10^{-3}	3.6×10^{-15}	3.6×10^{-15}
5	1.40598×10^{-4}	1.51773×10^{-4}	2.3×10^{-14}	5.8×10^{-15}
6	1.01740×10^{-5}	1.11379×10^{-5}	9.2×10^{-14}	2.1×10^{-13}
7	6.37908×10^{-7}	7.13147×10^{-7}	3.9×10^{-13}	6.2×10^{-12}
8	3.58926×10^{-8}	4.07033×10^{-8}	1.3×10^{-10}	9.9×10^{-11}

2.3 Radial Degree Extensions and Arithmetic Limits

The sampled added degrees are $10, 12, 14, 16$ on both disks at $N = 4$. Degrees $9, 11, 13$ were not run; the resolved endpoint is 12, not 13. Table 11 contains every completed field and both pre-assembly failures. Table 12 gives the independent field-discrepancy fractions; all three norms must be at most 1%. Assembly boosts are $16, 24$ and independent error boosts are $24, 32$. The squared-radial Duffy rule addresses the finite-regularity load; Gauss–Jacobi integration independently checks the exact-gradient integral. Paired error integration must agree within 10^{-4}. All successful integration repeats pass, including the flagged $p = 14$ fields; they do not remove the assembly-field disagreement. The initial run stopped during $p = 12$ on B; its completed A result and earlier degrees were retained and B was subsequently completed separately. The $p = 16$ failures occur in binary64 bubble-Gram Cholesky before assembly; there is no computed PDE field at that degree. Exact-rational Grams with 80/100-digit Cholesky succeed through $p = 16$, with pivot agreement within 1.1×10^{-64}. The `radial` and `diagnostics` commands regenerate the failure records and independent Gram diagnostics in their output directories.

Table 11: Every radial extension attempt. Flagged values are retained but excluded from the fits and shown open in main-paper Figure 9. A dash means no assembled PDE result.

Disk	p	L^2	Gradient	Full energy	Status
A	10	4.00992×10^{-5}	1.01136×10^{-3}	3.03783×10^{-3}	resolved
B	10	3.82520×10^{-5}	9.84956×10^{-4}	2.81065×10^{-3}	resolved
A	12	2.18939×10^{-5}	6.40970×10^{-4}	1.87526×10^{-3}	resolved
B	12	2.07672×10^{-5}	6.22347×10^{-4}	1.67274×10^{-3}	resolved
A	14	1.30170×10^{-5}	4.37636×10^{-4}	1.30376×10^{-3}	flagged
B	14	1.24337×10^{-5}	4.25449×10^{-4}	1.13333×10^{-3}	flagged
A	16	–	–	–	basis failed
B	16	–	–	–	basis failed

Table 12: Radial validation: field discrepancies relative to the smaller discretization error, and maximum relative change from independent error integration. The field limit is 1%; the integration limit is 10^{-4}.

Disk	p	L^2 (%)	Gradient (%)	Energy (%)	Max. integration repeat
A	10	0.00019	0.00021	0.00010	3.87×10^{-11}
B	10	0.00013	0.00015	0.00008	1.53×10^{-14}
A	12	0.04060	0.04800	0.02401	2.96×10^{-11}
B	12	0.02035	0.02440	0.01259	3.30×10^{-14}
A	14	4.93290	6.07673	2.93807	1.91×10^{-11}
B	14	1.23145	1.54998	1.02832	1.11×10^{-12}

The main-paper fits use exactly $p = 4, 5, 6, 7, 8, 10, 12$, with no inferred values at missing degrees. On A/B the slopes are $-3.33346/-3.35808$ in L^2, $-2.55778/-2.57543$ in gradient and $-2.59012/-2.61215$ in full energy. The $p = 2, \ldots, 8$ points remain in `radial-cases.csv`; the added attempts are in `radial-extension.csv`. Each `radial` run writes full validation records to `output/radial-results.json`. The older fits in Section 1.6 describe only their original degree window. The six extended fits are exported in `radial-fits.csv`, window `p4-12`.

2.4 Stabilization and Coefficient Conditioning

Table 13 uses saved matrices on the same smallest axis cut at $N = 6$, $p = 6$, $\epsilon = 3 \times 10^{-5}$ and $\beta_0 = 16$. Only the stabilization layer count and weight change. For each configuration, both assembly boosts $8, 12$ are checked. The Jacobi condition numbers agree within 2.66×10^{-10} relatively, endpoint eigenpair residuals are below 2.2×10^{-9} relative to their eigenvalues, and the energy-metric whitening errors are below 1.3×10^{-10}. Coercivity uses the norm with the same stabilization weight as the operator. No PDE field was recomputed, so this control makes no new solution-accuracy claim for the rescaled methods. Stronger or deeper stabilization improves conditioning on this sample, but does not close the gap to the fitted $\kappa_J = 132.6$ at the same degree. The `conditioning` command regenerates the paired records in `output/results.json`.

Table 13: Cached-matrix control of stabilization strength and depth. Reduction is relative to $L = 2$, $\gamma_1 = 0.1$; all four paired-rule checks pass.

L	γ_1	κ_J	Reduction	Coercivity
2	0.1	4.6503×10^6	1.00	0.949494
2	1.0	9.2873×10^5	5.01	0.963396
4	0.1	5.0481×10^5	9.21	0.966937
4	1.0	3.1452×10^5	14.79	0.975161

A Additional Numerical Details

A.1 Finite Regularity Energy Components

Figure 8 shows the separate weighted components from the same fourteen checked configurations. Their squares add to squared energy.

A.2 Schwarz Condition Estimates and Iteration Counts

These details use the same saved operators and stopping criteria as the main text.

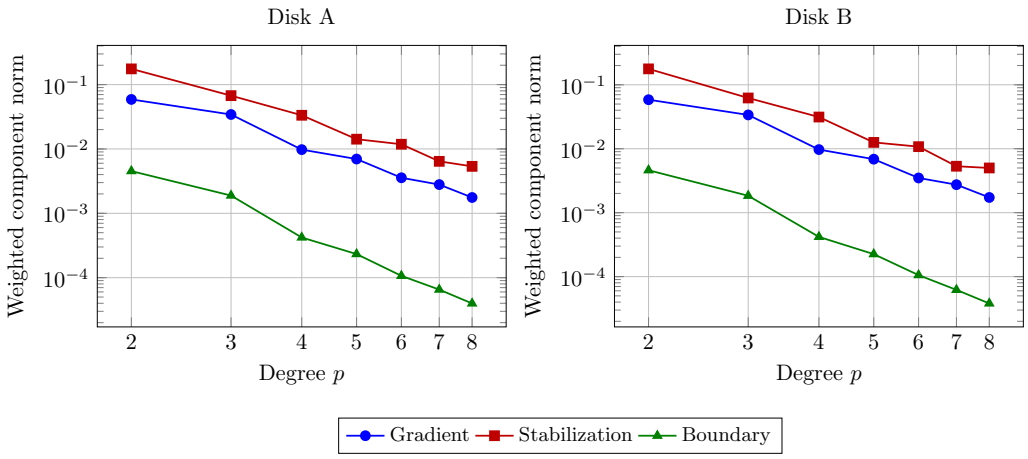


Figure 8: Finite-regularity energy contributions on the two disks, including the projection residual of the manufactured extension on complete patches.

Table 14: Smallest-offset cut operators: checked condition estimates and field-validated PCG iterations. The stabilization depth remains $L = 2$; subscript 4 denotes $L_{\mathrm{pc}} = 4$.

Path	p	κ_J	κ_4	PCG J	PCG $L_{\mathrm{pc}} = 4$
axis	2	2.01×10^2	3.68×10^0	88	16
axis	4	2.86×10^4	2.38×10^1	570	33
axis	6	4.65×10^6	3.57×10^3	2357	125
diagonal	2	2.78×10^2	3.20×10^0	67	14
diagonal	4	4.46×10^4	5.34×10^1	494	32
diagonal	6	5.07×10^6	3.99×10^4	2973	141

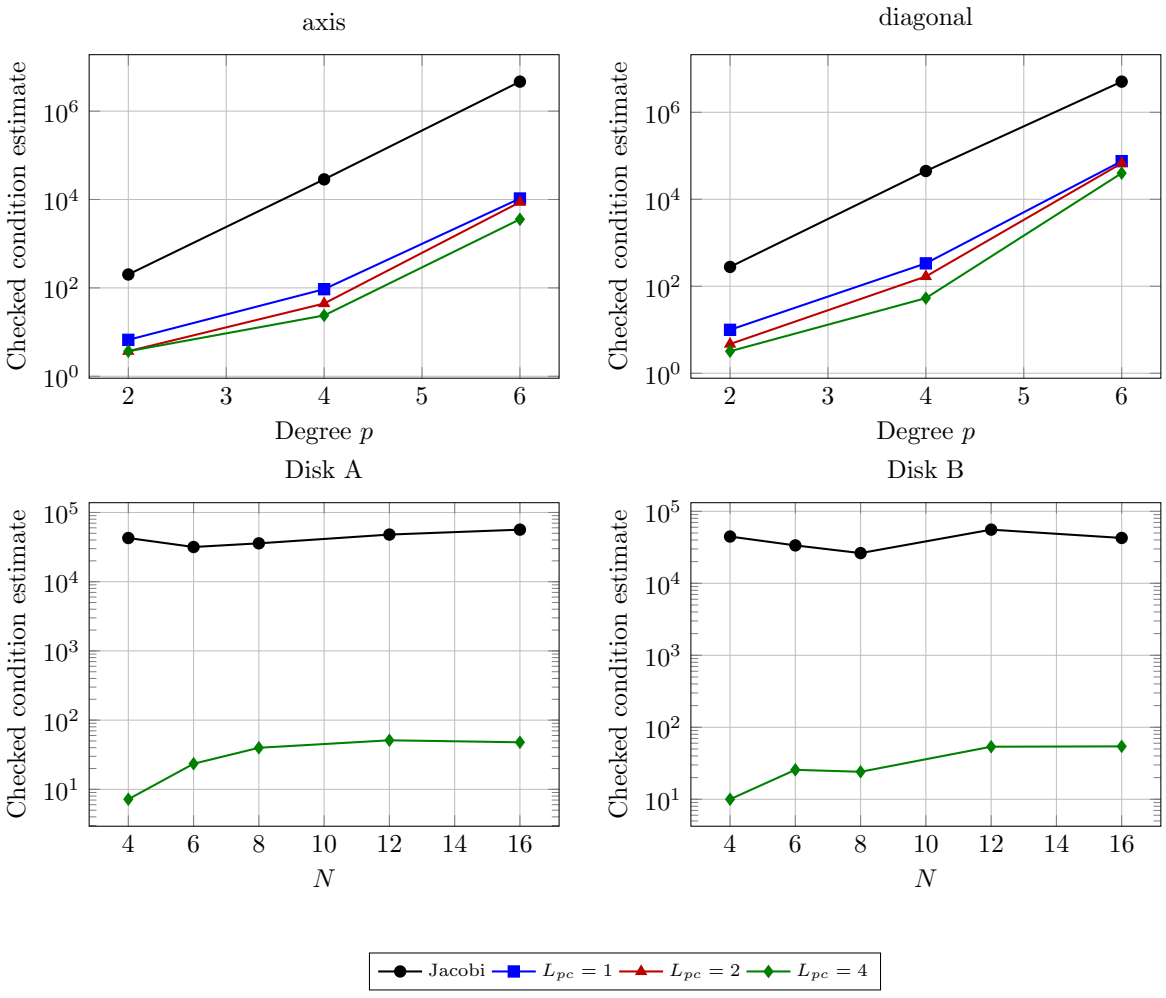


Figure 9: Checked preconditioned condition estimates versus degree on the two smallest-offset cuts (top) and mesh resolution on the fixed disks at degree four (bottom).

A.3 Higher-Degree Component Checks and Retained Flags

All accurate-field and reference norm repeats use the unchanged 10^{-4} tolerance. At degree twelve, three stabilization squared-component repeats for ordinary discrepancy fields exceed that tolerance after normalization by the reference squared energy error. Table 15 retains them; the complete comparison is not declared wholly accepted.

Table 15: Retained degree-twelve ordinary stabilization-component repeat flags. The reference-scale ratios exceed the declared 10^{-4} limit. Own-component ratios explain the scale but do not replace the criterion.

Ordinary discrepancy field	Reference-scale repeat	Own-component repeat
Boost 8 minus reference	5.368×10^{-2}	1.722×10^{-11}
Boost 12 minus reference	2.717×10^{-2}	9.480×10^{-12}
Boost 8 minus boost 12	1.820×10^{-2}	3.117×10^{-12}

The corresponding absolute squared-component changes are 9.92×10^{-28}, 5.02×10^{-28} and 3.36×10^{-28}. The maximum accurate/reference/load-series norm or component repeat is 9.22×10^{-15} at degree twelve. The ordinary and collapsed-coefficient accuracy failures remain in the saved criteria. Assembly, solve, integration and online-probe evidence are separate; no acceptance threshold is relaxed. The reproducibility records retain the failed comparison-conversion attempt and its recovery from completed degree-ten calculations.

A.4 Extended Mesh Measurements and Accuracy Flags

The additional sixteen configurations use the standard online binary64 field, assembly boosts 8/12 and error boosts 16/20. Table 16 retains the finest-mesh errors even when the accuracy checks fail. Passing integration repeats do not resolve the disagreement between the two assembled fields, and the new configurations have no independent reference solve.

Table 16: Measured online errors at $N = 24$, assembly boost 12 and error boost 20. Daggers retain values flagged in all three norms; they are excluded from validated fits.

Disk	p	L^2	Gradient	Energy
A	2	3.068×10^{-5}	1.939×10^{-3}	3.300×10^{-3}
A	4	7.611×10^{-9}	5.954×10^{-7}	2.488×10^{-6}
A	$6^{\dagger}$	2.640×10^{-12}	2.213×10^{-10}	9.234×10^{-10}
A	$8^{\dagger}$	1.060×10^{-12}	1.746×10^{-10}	3.247×10^{-10}
B	2	2.770×10^{-5}	1.813×10^{-3}	3.346×10^{-3}
B	4	7.578×10^{-9}	5.727×10^{-7}	2.571×10^{-6}
B	$6^{\dagger}$	2.792×10^{-12}	2.324×10^{-10}	9.587×10^{-10}
B	$8^{\dagger}$	1.306×10^{-12}	2.067×10^{-10}	3.726×10^{-10}

Table 17 lists the field discrepancies, and Figure 10 shows the corresponding energy components. The added degree-eight $N = 4$ configurations pass every declared check on both disks. The original $N = 6, p = 8$ physical-norm reference flags and the separate degree-twelve component-repeat flags above remain unchanged. Neither paired assembly agreement nor integration agreement excludes a shared bias.

Table 17: All ten newly flagged configurations: assembly-field differences divided by the smaller measured error, in percent. Every entry exceeds the unchanged 1% limit. Norm integration and component-repeat tolerances remain 10^{-4}. Additional last-correction solve flags are retained in the criteria data.

Disk	p	N	L^2 (%)	Gradient (%)	Energy (%)
A	6	24	3.12	4.38	1.85
A	8	8	10.14	14.32	6.07
A	8	12	93.70	107.04	101.45
A	8	16	206.15	196.27	190.68
A	8	24	121.12	126.69	142.27
B	6	24	3.41	5.48	2.25
B	8	8	18.53	23.80	8.88
B	8	12	205.10	224.06	155.22
B	8	16	247.62	228.14	220.20
B	8	24	181.98	170.01	174.81

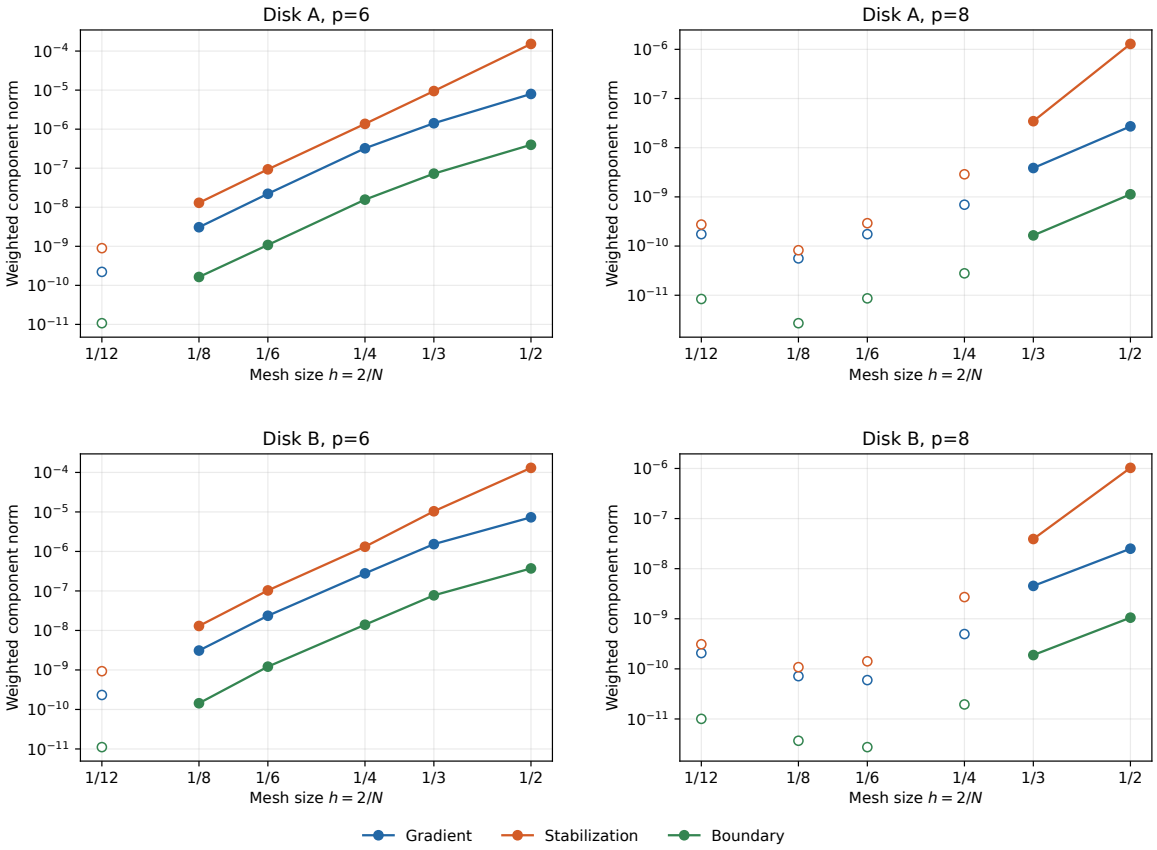


Figure 10: Weighted energy-component square roots for the extended degree-six and degree-eight mesh families on disks A and B. Their squares sum to squared energy. Open markers retain configurations whose energy checks fail; connected curves do not bridge them.

Physical Energy Components. Table 18 separates the physical part (1.2) from full energy for selected resolved smooth cases. Stabilization contributes 97.8–97.9% at $N = 6, p = 6$, but only 65.3–70.5% at $N = 24, p = 2$. Thus a single "95%" description is not uniform across the study. All 118 component decompositions are supplied in the accompanying data; derived physical parts do not introduce new independent accuracy certificates or change any saved acceptance flag.

Table 18: Physical and full energy errors for four resolved smooth cases. The final column is the stabilization fraction of squared full energy.

Disk	N	p	E_{phys}	$\|\|\|Eu - u_h\|\|\|_h$	Stab. share
A	6	6	1.420×10^{-6}	9.567×10^{-6}	97.8%
B	6	6	1.533×10^{-6}	1.051×10^{-5}	97.9%
A	24	2	1.945×10^{-3}	3.300×10^{-3}	65.3%
B	24	2	1.818×10^{-3}	3.346×10^{-3}	70.5%

A.5 Exact Tube Certificates

The geometric question is whether the existing patch family admits the partition required by (2.9), not whether each boundary element's own seed patch contains its tube. A first sufficient test excluded patch-boundary edges from whole annular sectors. That test certified only 8 of the 30 seed-based configurations and retained eleven explicit violations. It can also fail when a ray merely touches a patch vertex or different parts of an arc require different existing patches. The `tube_h_over_16_certified` column in the 78-row `tube-summary` seed audit records this conservative sufficient test, not the exact seed-containment result in Table 19. At the saved patch depths, its eight passes comprise two two-layer and six four-layer geometries. The exact crossing test gives six seed passes in each group. For example, `geometry-01` fails the conservative test but passes the exact seed test without reassignment. A false value in that column is therefore not by itself a certificate that the seed tube leaves its patch. We use the exact crossing arrangement described next. In the main paper's notation, the assigned boundary pieces form $\Gamma_{\mathcal{P}}$, and the certified tubes $U_{\mathcal{P}}$ have depth $c_\delta h$ with $c_\delta = 1/16$.

Complete Angular Partition. For a disk of radius R, set $r_0 = R - h/16$. The open triangles traversed by a radial segment from radius R to radius r_0 can change only when one of its endpoints crosses a mesh edge, or the segment passes through a mesh vertex in the closed annulus $r_0 \le |x - c| \le R$. We include all intersections of both circles with mesh supporting lines and every radial direction of such a vertex. Supporting lines may introduce extra events, which only subdivide the partition. Collinearity with a mesh edge is already included among its endpoint events. Between consecutive events, the traversed-triangle set is constant: crossings vary continuously and their order cannot change without a vertex or endpoint event.

On each quadrant, the rational parametrization $n(u) = ((1 - u^2)/(1 + u^2), 2u/(1 + u^2))$, rotated as appropriate, turns every event equation into a quadratic with rational coefficients. All real roots are isolated with brackets verified by exact signs; rational roots are retained exactly. A rational parameter strictly between consecutive brackets supplies a representative segment. Exact segment–triangle intersections determine its complete traversal. Absence of an intervening event extends this traversal to the entire open angular piece, including the portions adjacent to the root brackets. Thus this is a continuum certificate, not an inference from sampled normal rays. The finite set of event angles has zero boundary measure.

Assignment and Verification. Keep the seed patch if it contains every traversed triangle. Otherwise, among existing patches containing the complete boundary element and the traversal,

choose the smallest patch, breaking ties by seed index. All 5324 angular pieces admit an assignment; 97 pieces on 18 geometries are reassigned. Table 19 summarizes the result. The patch elements, overlaps, projections and stabilization terms are unchanged. By Remark 4.7, the discrete matrices, loads, energy norm and saved PDE fields are therefore unchanged.

Table 19: Exact tube certification at depth $h/16$ within the unchanged patch families. Pass counts refer to complete geometries; the final column counts open angular pieces, not PDE configurations.

Saved patches	Geometries	Seed passes	Assigned passes	Reassigned pieces
Two layers	24	6	24	97
Four layers	6	6	6	0

An independent verifier reconstructs the exhaustive event-polynomial set, checks the number and isolation of all roots, and verifies each supplied segment interval by triangle half-plane inequalities at both endpoints. Convexity proves each whole interval lies in its triangle; their union covers the complete segment. It also checks the selected patch membership and all eleven earlier counterexample points in their newly assigned patches. This verifier does not call the search, root finder or segment-clipping routines. The input coordinates are the exact real values of the retained binary64 data. The certificate does not cover unspecified nearby geometries or establish a uniform algorithm for untested cuts.

The original obstruction remains instructive. With a grid vertex at $v = 0$, southwest-to-northeast diagonals, $n = (3/5, 4/5)$ and disk center $c = -(R - \epsilon h)n$, the boundary point $\epsilon h n$ lies in the northeast upper triangle. Its inward ray reaches the southwest lower triangle after depth ϵh. Those triangles have face-adjacency distance three, so the natural two-layer seed patch misses the latter. This rules out a cut-uniform depth for that seed assignment, but a neighboring seed patch can contain both triangles. The certified partition uses this freedom. Since it succeeds for all retained cases, patch enlargement and a replacement PDE campaign are unnecessary for this geometric issue. Existing numerical accuracy flags and the separate questions of general cut uniformity and corner regularity remain.

B Hierarchical Triangular Basis

The numerical space uses a conforming vertex–edge–bubble basis on each affine triangle. Its vertex and edge functions belong to the hierarchical H^1 construction of Schöberl and Zaglmayr [1, Section II]. The interior functions use a separate reference H^1 orthogonalization. This is not the standard L^2-orthogonal Dubiner basis; only the bubble block is orthogonalized here.

Vertex and Edge Functions. On the reference triangle with vertices $(0,0), (1,0), (0,1)$, let $\lambda_1 = 1-r-s$, $\lambda_2 = r$, and $\lambda_3 = s$. The three vertex functions are λ_i. Write P_k for the Legendre polynomial normalized by $P_k(1) = 1$, and define

$$L_n(t) = \int_{-1}^{t} P_{n-1}(z)\, dz \qquad n \geq 2 \tag{B.1}$$

Orient every edge from its smaller to its larger global vertex index. For the corresponding local vertices i, j, its modes are

$$\phi_{ij,n} = (\lambda_i + \lambda_j)^n L_n\left(\frac{\lambda_j - \lambda_i}{\lambda_i + \lambda_j}\right) \qquad n = 2, \ldots, p \tag{B.2}$$

The right side denotes its polynomial extension at $\lambda_i + \lambda_j = 0$. Since $L_n(\pm 1) = 0$, each mode contains the factor $\lambda_i \lambda_j$ and vanishes on the other two edges. On its own edge it restricts to L_n, with the same orientation from either adjacent element.

Interior Functions. For $p \geq 3$, start from the linearly independent bubbles

$$b_{ab}(r,s) = \lambda_1\lambda_2\lambda_3 r^a s^b \qquad a,b \geq 0, \quad a+b \leq p-3 \tag{B.3}$$

Enumerate them by increasing total degree and assemble the reference Gram matrix

$$G_{k\ell} = (b_k, b_\ell)_{\widehat{T}} + (\nabla b_k, \nabla b_\ell)_{\widehat{T}} \tag{B.4}$$

In exact arithmetic G is positive definite. If $G = CC^T$ is its Cholesky factorization, define

$$\widetilde{b}_k = \sum_\ell (C^{-1})_{k\ell} b_\ell \tag{B.5}$$

These functions are orthonormal within the bubble space for the full reference H^1 inner product. They are not asserted to be orthogonal to the vertex and edge functions. The bubble block is empty for $p < 3$.

Spanning and Conformity. After subtracting its vertex interpolant, any $q \in \mathbb{P}_p(\widehat{T})$ has edge traces that vanish at both endpoints. The integrated-Legendre traces span these univariate polynomials, so the edge functions can remove them independently. The remaining polynomial vanishes on all three edges and is therefore divisible by $\lambda_1\lambda_2\lambda_3$, with quotient in $\mathbb{P}_{p-3}$. It lies in the bubble span, which the invertible Cholesky transformation preserves. The count is

$$3 + 3(p-1) + \frac{(p-1)(p-2)}{2} = \frac{(p+1)(p+2)}{2} = \dim \mathbb{P}_p(\widehat{T}) \tag{B.6}$$

Thus the functions form a basis in exact arithmetic. Consistent edge orientation identifies identical traces on adjacent affine triangles; the bubbles have zero trace. Assembly therefore gives the conforming $\mathbb{P}_p$ space used in the main paper.

Stored Polynomials and Arithmetic Checks. The implementation constructs edge coefficients in exact rational arithmetic and retains explicit edge factors. After the numerical bubble Gram factorization, it reconstructs bubble quotients and stores the product with $rs(1-r-s)$ explicitly, preserving zero edge traces rather than relying on cancellation of rounded expanded coefficients. This preserves the polynomial space; finite-precision coefficients need not retain exact reference orthogonality.

The basis checks include exact polynomial equality of shared-edge traces, exact zero bubble traces, the edge derivative Gram identities, linear reproduction, and sampled values and gradients compared with independent higher-precision polynomial evaluation. The spanning argument above is separate from these implementation checks. Neither implies uniform conditioning in p: the retained binary64 bubble-Gram Cholesky failure at $p = 16$, and the separate field-accuracy flags, remain part of the reported evidence.

References

[1] J. Schöberl and S. Zaglmayr. High order Nédélec elements with local complete sequence properties. *COMPEL - The International Journal for Computation and Mathematics in Electrical and Electronic Engineering*, 24(2):374–384, 2005. doi:10.1108/03321640510586015.